

System Documentation

Electronic control system

BlueLine

with multi-function display

Series 4000 Waterjet

MDEC

Application: Marine

Functional Description

Operating Instructions

Workshop Manual

Installation and

Commissioning Instructions

E532780/03E



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Table of Contents

1	Safety		4.1.2.8	Switching off the overall BlueLine system	77
1.1	Important requirements for all products	6	4.1.2.9	PPC - Controls and indicators	78
1.2	Personnel and organizational requirements	7	4.1.2.10	Priming Pump Controller PPC - Function check	79
1.3	Safety requirements for startup and operation	8	4.2	Engine Control System ECS-5	80
1.4	Safety requirements for maintenance and repair work	9	4.2.1	Controls and Displays	80
1.5	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials	12	4.2.1.1	Local Operating Panel LOP	80
1.6	Standards for safety notices in the text	14	4.2.1.2	Local Operating Panel LOP with display	81
2	General Information		4.2.2	Normal Operation: LOS or LOP with Display	83
2.1	Engine side and cylinder designations	15	4.2.2.1	Engine - Switching ready for operation	83
3	Functional Description		4.2.2.2	Activating Local mode	84
3.1	Product Summary	16	4.2.2.3	Engine - Start	85
3.1.1	BlueLine system - Overview	16	4.2.2.4	Local mode - Gearbox engagement/disengagement	86
3.1.2	BlueLine subsystems - Use	19	4.2.2.5	Local mode - Engine speed adjustment	87
3.2	Subsystems	20	4.2.2.6	Stopping the engine	88
3.2.1	Monitoring and Control System MCS-5	20	4.2.2.7	Deactivating Local mode	89
3.2.1.1	MCS-5 BlueLine - Overview	20	4.2.2.8	Engine start - Interlock	90
3.2.1.2	MCS-5 BlueLine - Use of the devices	22	4.2.2.9	Engine - Emergency shutdown	91
3.2.1.3	RS422 PIM 4 serial interface as per MTU specifications	25	4.2.3	Operation during Servicing: LOP without Display	92
3.2.1.4	MCS-5 BlueLine - Main control stand	30	4.2.3.1	Engine - Starting	92
3.2.1.5	MCS-5 BlueLine - Slave control stand	33	4.2.3.2	Stopping the engine	93
3.2.2	Engine Control System ECS-5	34	4.2.3.3	Interlocking engine start	94
3.2.2.1	ECS-5 BlueLine - Overview	34	4.2.3.4	Engine - Emergency shutdown	95
3.2.2.2	ECS-5 BlueLine - Use of the devices	38	4.3	Troubleshooting	96
3.2.3	Remote Control System	42	4.3.1	Faults in the BlueLine system	96
3.2.3.1	Interface to remote control system	42	4.3.2	Troubleshooting	98
3.2.3.2	Remote control interface - Use of devices	44	4.3.3	Alarms - General information	101
4	Operating Instructions		4.3.4	Single-point alarms - Acknowledgment	102
4.1	Monitoring and Control System MCS-5	45	4.3.5	Fault indication on the display of LOP with display DIS	103
4.1.1	Controls and Displays	45	4.3.6	Fault indications on the MFD display of the main control stand	108
4.1.1.1	MCS-5 - Controls and displays	45	4.3.7	Fault indication on printed circuit board MPU 29	114
4.1.1.2	LOP display - Overview of screen pages	49	4.3.8	Fault messages - Engine Control Unit	118
4.1.1.3	Display MFD - Overview of screen pages	59	4.3.9	Oil priming pump PPC - Troubleshooting	165
4.1.2	Operation	70	4.4	Task Description	166
4.1.2.1	Switching on the overall BlueLine system	70	4.4.1	Cabling	166
4.1.2.2	Brightness of indicators/instruments - Adjustment	71	4.4.1.1	Engine wiring - Check	166
4.1.2.3	Engine - Start	72	4.4.2	Control and Display Elements, General	167
4.1.2.4	Override function - Activation	73	4.4.2.1	Indicator lamp - LED replacement	167
4.1.2.5	Override function - Deactivation	74	5	Workshop Manual	
4.1.2.6	Stopping the engine	75	5.1	Task Description	169
4.1.2.7	Emergency engine shutdown	76	5.1.1	Operating Tests	169
			5.1.1.1	BlueLine test mode - Execution	169
			5.1.1.2	Overspeed test	171
			5.1.1.3	Emergency stop test	172
			5.1.1.4	Cranking	173

5.1.2	Checks and Settings	174
5.1.2.1	LOP display DIS settings - Check	174
5.1.2.2	LOP display DIS - Setting	175
5.1.2.3	MFD settings - Check	176
5.1.2.4	MFD display - Set	177
5.1.2.5	Battery on MFD display - Replacement	179
5.1.2.6	PIM 4 node number - Check	181
5.1.2.7	PIM 4 node number - Setting	182
5.1.2.8	Serial interface terminator plugs - Check	183
5.1.2.9	Emergency stop shorting plug X18 - Check	184
5.1.2.10	Jumper configuration of MCS-5 control unit - Check	186
5.1.2.11	CAN bus connections PIM 4 - Check	187
5.1.2.12	CAN bus line resistance - Check	188
5.1.2.13	MPU 29 node address in LOP - Check	190
5.1.2.14	MPU 29 node address in LOP - Setting	191
5.1.2.15	Control stand configuration in Local Operating Panel LOP - Check	192
5.1.2.16	Shaft number setting - Check	194
5.1.2.17	LOP power supply - Check	195
5.1.2.18	Jumper configuration of LOP - Check	198
5.1.2.19	GCU 6 jumper configuration - Check	199
5.1.2.20	GCU node number - Check	200
5.1.2.21	BlueLine CAN node configuration	201
5.1.3	Repair Work	203
5.1.3.1	PIM 4 cover - Removal	203
5.1.3.2	PIM 4 cover - Installation	204
5.1.3.3	Analog display instrument VDO OceanLine - Replacement	205
5.1.3.4	Lamp in analog display instrument VDO OceanLine - Replacement	206
5.1.3.5	Indicator lamp - Replacement	207
5.1.3.6	Indicator lamp - Removal	208
5.1.3.7	Indicator lamp - Installation	209
5.1.3.8	Indicator lamp - LED replacement	210
5.1.3.9	Contactors - Replacement	212
5.1.3.10	LED in pushbutton/switch - Replacement	213
5.1.3.11	MFD display - Replace	215
5.1.3.12	SD card on MFD display - Replacement	216
5.1.3.13	PIM 4 parameters - Reset	217
5.1.3.14	Fuse in control unit PIM 4 - Replacement	218
5.1.3.15	Control unit PIM 4 - Replacement	219
5.1.3.16	Printed circuit board in module cassette - Replacement	221
5.1.3.17	Fuse on printed circuit board in PIM cassette - Replacement	223
5.1.3.18	Printed circuit board MPU 29 - Removal	225
5.1.3.19	Printed circuit board MPU 29 - Installation	226
5.1.3.20	Printed circuit board MPU 29 - Replacement	227
5.1.3.21	Data module MEM on MPU 29 - Replacement	228
5.1.3.22	Data module MEM on MPU 29 - Transfer	229
5.1.3.23	LOP fuse - Replacement	230
5.1.3.24	LOP with display front flap - Replacement	231
5.1.3.25	LOP - Replacement	233
5.1.3.26	Gear Control Unit GCU - Replacement	234
5.1.3.27	Priming Pump Controller PPC - Repair	236

5.1.3.28	Priming Pump Controller PPC - Replacement	238
5.1.4	Supplementary Technical Information	239
5.1.4.1	Control unit PIM 4 MCS - Internal design	239
5.1.4.2	Local Operating Panel LOP - Design	245
5.1.4.3	Gear Control Unit GCU 6 - Design	253
5.1.4.4	Priming pump control unit PPC design	258

6 Installation and Commissioning Instructions

6.1	Installation	259
6.1.1	Preparation	259
6.1.1.1	Cable routes and openings between all installation locations - Check	259
6.1.1.2	Cables between installation locations - Routing	260
6.1.2	Mechanical Installation	262
6.1.2.1	PIM 4 module housing - Installation	262
6.1.2.2	OceanLine analog display instruments - Installation	263
6.1.2.3	Horn VDO - Installation	265
6.1.2.4	Indicator lamp - Installation	266
6.1.2.5	Illuminated pushbuttons/key switches - Installation	267
6.1.2.6	GCU - Installation	269
6.1.2.7	Priming Pump Controller PPC - Installation	270
6.1.2.8	Display MFD - Installation	271
6.1.2.9	LOP - Installation	272
6.1.2.10	LOP with display - Conversion to application with remote control system (shipyard)	274
6.1.2.11	Terminal box XO10 - Installation	276
6.1.3	Electrical Installation	277
6.1.3.1	HSK cable gland - Preparation	277
6.1.3.2	Terminal block - Preparation	278
6.1.3.3	PIM 4 cover - Removal	279
6.1.3.4	GCU 6 - Connection	280
6.1.3.5	Waterjet remote control system (shipyard) - Connection	282
6.1.3.6	Non-MTU gearbox - Connection	286
6.1.3.7	Display instruments on main control stand - Connection	289
6.1.3.8	Controls and status indicator lamps on main control stands - Connection	290
6.1.3.9	Indicator lamps for single-point alarms - Connection	291
6.1.3.10	Devices on a slave control stand - Connection	292
6.1.3.11	PIM 4 MCS control unit - Connection	293
6.1.3.12	PPC - Connection (single phase)	296
6.1.3.13	PPC - Connection (three phases)	298
6.1.3.14	Starter and battery-charging generator - Connection	300
6.1.3.15	Engine Control Unit ECU - Connection	305
6.1.3.16	Barring tool limit switch - Connection	306
6.1.3.17	Display MFD - Connection	307
6.1.3.18	LOP - Connection	309

6.1.3.19	Additional Yard signals (optional) - Connection	312	6.2.2	Settings	333
6.1.3.20	Terminal box X010 - Connection (battery-charging generator charges starter battery)	314	6.2.2.1	BlueLine CAN node configuration	333
6.1.4	Data Transmission	317	6.2.2.2	ECU configuration	335
6.1.4.1	RS422 - Transmission values	317	6.2.2.3	Minidialog - Settings	336
6.1.5	Technical Data	318	6.2.3	Tests	349
6.1.5.1	Technical data - LOP	318	6.2.3.1	Initial engine start-up at the service interface of the LOP	349
6.1.5.2	Display MFD - Technical data	319	6.2.3.2	Initial engine startup at the display of Local Operating Panel LOP	350
6.1.5.3	ECU 4 - Technical data	321	6.2.3.3	Initial engine start at MCS	351
6.1.5.4	GCU 6 - Technical data	322			
6.1.5.5	PPC - Technical data	323	7	Appendix A	
6.1.5.6	Technical data X010	324	7.1	Abbreviations	352
6.1.5.7	PIM 4 control unit - Technical data	325	7.2	Conversion tables	357
6.2	Initial Operation	326	7.3	MTU contact persons/service partners	361
6.2.1	Preparation	326	8	Appendix B	
6.2.1.1	Checks prior to startup	326	8.1	Special Tools	362
6.2.1.2	Operating voltage - Initial application	327	8.2	Index	363
6.2.1.3	Supply voltage distribution - Check	329			
6.2.1.4	Data module MEM on MPU 29 - Installation	331			
6.2.1.5	SD memory card in display MFD - Installation	332			

1 Safety

1.1 Important requirements for all products

Nameplate

The product is identified by a nameplate, model designation or serial number which must match the information on the title page of this manual.

Nameplate, model designation or serial number can be found on the product.

General information

This product may pose a risk of injury or damage in the following cases:

- Improper use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications
- Noncompliance with the safety instructions and warning notices

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (including engine management/parameters)
- In compliance with all safety regulations and in accordance with all warning notices in this manual
- With maintenance work performed in accordance with the (→ Maintenance Schedule) throughout the useful life of the product
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair
- By contracting only workshops authorized by the manufacturer to carry out repair and overhaul

Any other use shall be considered non-intended. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

Changes or modifications

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

1.2 Personnel and organizational requirements

Organizational measures of the operator

This manual must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use this manual as a basis for instructing personnel on product operation and repair, whereby the safety-relevant instructions, in particular, must be read and understood.

This is particularly important in the case of personnel who only occasionally perform work on or around the product. This personnel must be instructed repeatedly.

Personnel requirements

All work on the product shall be carried out by trained and qualified personnel only.

- Training at the Training Center of the manufacturer
- Qualified personnel specialized in mechanical and plant engineering

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair and transport.

Working clothes and personal protective equipment

Wear proper protective clothing for all work.

When working, always wear the necessary personal protective equipment (e.g. ear protectors, protective gloves, goggles, breathing protection). Observe the information on personal protective equipment in the respective activity description.

1.3 Safety requirements for startup and operation

Safety requirements for startup

The product must be installed and accepted in accordance with manufacturers' specifications prior to initial startup.

All the requisite regulatory approvals must have been granted and all startup requirements fulfilled prior to initial startup.

Whenever the product is started, always ensure that:

- All maintenance and repair work has been completed.
- All loose parts have been removed from rotating machine components.
- Wearers of cardiac pacemakers or other active medical implants are well clear.
- The operating room is adequately ventilated.
- Exhaust pipework is leak-tight and routed to atmosphere.
- Battery terminals, generator terminals and cables are guarded to preclude accidental contact.
- All personnel is clear of the danger zone represented by moving parts.

Immediately after putting the product into operation, make sure that all control and display instruments as well as the monitoring, signaling and alarm systems are working properly.

Safety requirements for operation

The operator must be familiar with the controls and displays.

The operator must be aware of the consequences of any operations he/she performs.

During operation, the display instruments and monitoring units must be constantly observed in regard of present operating status, limit value violation and warning or alarm messages.

Malfunctions and emergency stop

Emergency procedures, in particular emergency stop, must be practiced on a regular basis.

Take the following steps if a system malfunction is detected or signaled by the system:

- Inform the duty supervisor(s).
- Evaluate the message.
- Respond to the emergency appropriately, e.g. execute an emergency stop.

Operation

Do not stay in the operating room when the product is running unless absolutely necessary and then as briefly as possible.

Keep a safe distance away from the product whenever possible. Do not touch the product unless expressly instructed to do so.

Do not inhale exhaust gases emitted by the product.

Ensure that the following requirements have been fulfilled before starting the product:

- Ear protectors are worn.
- Mop up any leaked or spilled fluids and lubricants immediately or soak up with a suitable binding agent.

Operation of electrical equipment

Some components are live (high voltage) when electrical equipment is in operation.

Observe the safety instructions for these appliances.

1.4 Safety requirements for maintenance and repair work

Safety requirements before commencing maintenance and repair work

Have maintenance or repair work carried out by qualified and authorized personnel only.

Allow the product to cool down to less than 50°C (risk of explosion for oil vapors, fluids and lubricants, risk of burning).

Relieve pressure in fluid and lubricant systems and compressed-air lines which are to be opened. Use suitable collection vessels with a sufficient filling volume.

Ensure that the operating room is adequately ventilated when changing the oil or working on the fuel system.

Do not perform maintenance or repair work when the product is running unless:

- expressly instructed to do so.
- the product is running in the low load range and only for as long as necessary to complete the task.

Lock-out the product to preclude undesired starting, e.g. start interlock.

Tag-out the product with a “Do Not Start” sign in the operation room or on the control facility.

Disconnect the battery. Lock the contactor.

Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.

Disconnect the control facility from the product.

For starters with pinions made of copper-beryllium alloy:

- Wear a respirator mask (filter class P3). Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the interior of the flywheel housing with a class H dust extractor.
- Observe the safety data sheet.

Safety requirements during maintenance and repair work

Take special care when removing vent plugs or plug screws from the product. Hold a cloth over the screw or plug to prevent discharge of highly pressurized liquids.

Take care when draining hot fluids and lubricants (risk of burning).

Use suitable and calibrated tools only. Observe the specified tightening torques during assembly or disassembly.

Carry out work only on assemblies and/or installations which are properly secured.

Never climb up on the lines.

Keep fuel injection lines and connections clean.

Always seal connections with caps or covers if a line is removed or opened.

Fit new seals when re-installing lines.

Avoid damaging lines, particularly the fuel lines.

Ensure that all retainers and dampers are installed correctly.

Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Never route fuel or oil lines in the vicinity of hot components.

Do not touch elastomeric seals (e.g. Viton sealing rings) with your bare hands if they have a carbonized or resinous appearance.

Observe the cooling time for components which have been heated for installation or removal (risk of burning!).

Always use suitable ladders and work platforms when working above head-height. Ensure that components or assemblies are placed on stable surfaces.

Pay particular attention to cleanliness at all times.

Safety requirements after completing maintenance and repair work

Ensure that all personnel is clear of danger zones before cranking.

Check that all access ports/apertures which have been opened to facilitate working are closed again.

Check that all safety equipment has been installed and that all tools and loose parts have been removed (especially the barring gear).

Ensure that no unattached parts have been left in/on the product (e.g. including rags and cable straps).

Welding work

Do not perform welding on the product or its attachments. Cover the product when performing welding work in the vicinity.

Before commencing welding work:

- Switch off the master power supply switch.
- Disconnect the battery.
- Disconnect electronic and genset grounds.

Do not perform maintenance or repair work on the product when welding is in progress in its vicinity. Risk of explosion or fire due to oil vapors and highly flammable process materials.

Do not use the product as a grounding terminal.

Do not route the welding cable over or near the wiring harnesses of the product. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove components (e.g. exhaust pipe) from the product before performing necessary welding work .

Hydraulic installation and removal

Check function and satisfactory condition of the jigs and fixtures to be used. Use only the specified jigs and fixtures for hydraulic removal/installation.

Observe the max. permissible force-on pressure specified for the installation/removal jig.

Do not attempt to bend or exert force on HP lines.

Before starting work, pay attention to the following:

- Vent the installation/removal jig, the pumps and the pipework at the relevant designated points.
- For hydraulic installation, screw on the jig with the piston retracted.
- For hydraulic removal, screw on the jig with the piston extended.

For hydraulic installation/removal jigs with central expansion pressure supply, screw the spindle into the shaft end until correct sealing has been established.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

Working with batteries

Observe the safety requirements of the battery manufacturer when working with batteries.

Gases released from the battery are explosive. Avoid sparks and naked flames.

Do not allow electrolyte (battery acid) to come into contact with skin or clothing.

Wear goggles and protective gloves.

Do not place tools on the battery.

Check polarity before connecting the cable to the battery. Battery polarity reversal may lead to injury by the sudden discharge of acid or bursting of the battery unit.